

Line Side Board (LSB) Plant Deployment Work Instruction

Full installation & configuration procedure · Production environment

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Version: 1.0

Applies to: LSB v2.0 · Go + Java + React

Audience: IT / Plant Systems Administrator

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Before you begin

Work in an **elevated (Administrator) Command Prompt** for all service-related steps. Coordinate with your DBA for MARS SQL Server read-only credentials and PostgreSQL server details before arriving at the plant.

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1

Prerequisites & System Requirements

Verify everything is in place on the plant server before copying any files

1.1

Minimum server specification

Component	Minimum	Notes
OS	Windows Server 2019 or later	Windows 10/11 Pro also supported
CPU	4 cores	
RAM	8 GB	16 GB recommended
Disk	40 GB free	For app, logs, and PostgreSQL data

Network	Access to MARS MSSQL server	Port 1433 must be open
Network	Reachable from shop-floor PCs	Port 3001 (dashboard)

1.2

Required software — install in this order

Software	Minimum version	Where to get it	Badge
Java JRE	17	adoptium.net (Eclipse Temurin)	REQUIRED
PostgreSQL	15	postgresql.org	REQUIRED
Microsoft ODBC Driver 18 for SQL Server	18	Microsoft Download Center	REQUIRED
NSSM (Non-Sucking Service Manager)	2.24	nssm.cc — copy <code>nssm.exe</code> to <code>C:\Windows\System32\</code>	REQUIRED
Python 3 + pyodbc	3.9+	python.org , then <code>pip install pyodbc</code>	OPTIONAL — FOR DIAGNOSTICS

i After installing Java, verify it is on the system PATH: open a new Command Prompt and run `java -version`. You should see `openjdk 17.x.x` or similar. If not, add `C:\Program Files\Eclipse Adoptium\jdk-17.x.x\bin` to the system PATH environment variable.

1.3

PostgreSQL — create database and user

During PostgreSQL installation, set a strong *postgres* superuser password and note it. Then open **pgAdmin** (or `psql`) and run:

```
-- Run as postgres superuser
CREATE USER lsb_admin WITH PASSWORD 'choose-a-strong-password';
CREATE DATABASE lsb_oe OWNER lsb_admin;
GRANT ALL PRIVILEGES ON DATABASE lsb_oe TO lsb_admin;
```

⚠ Note the password you choose — you will enter it in the dashboard setup wizard in Phase 6.

1.4

Collect MARS connection details from your DBA

You will need the following before starting the setup wizard. Write them down or store them in your password manager:

Item	Example	Your value
------	---------	------------

MARS SQL Server hostname / IP	<code>mars-sql.plant01.local</code>
-------------------------------	-------------------------------------

MARS SQL Server port	<code>1433</code>
----------------------	-------------------

MARS database name	<code>MARS_DB</code>
--------------------	----------------------

Read-only username	<code>lsb_reader</code>
--------------------	-------------------------

Password	<i>(from DBA)</i>
----------	-------------------

**MARS access must be read-only.**

The LSB application never writes to MARS. Ask the DBA to grant only `SELECT` permission on the relevant tables. Do not use `sa` or any admin account.

2

Build Deliverables on the Development Machine

Compile the Go and Java binaries before copying to the plant server



This phase is done

on your development PC

— not on the plant server. You need Go 1.21+ and Maven/JDK 17+ installed on the dev machine. You do *not* need Go or Maven on the plant server.

2.1

Build the Go binary

Open a Command Prompt in the project root, then:

```
cd backend-go
set GOOS=windows
set GOARCH=amd64
go build -o lsb-api.exe .\exe\
```

Expected output: no errors, file `backend-go\lsb-api.exe` created (roughly 15–25 MB).

2.2

Build the Java JAR

```
cd backend-java
mvn clean package -DskipTests
```

Expected output: `BUILD SUCCESS`. File created: `backend-java\target\mars-service-2.0.0.jar` (roughly 50–80 MB including embedded Tomcat).

2.3

Verify both artifacts exist before packaging

```
dir backend-go\lsb-api.exe
dir backend-java\target\mars-service-2.0.0.jar
```

Both lines must show a file size. If either shows *File not found*, repeat the relevant build step.

2.4

Package the deployment folder

Copy the entire project folder to a USB drive or a network share the plant server can reach. The folder structure must include:

```
Drishti-LineSideBoard\
├─ backend-go\
│   ├── lsb-api.exe          <-- built in step 2.1
│   └─ db\schema.sql
├─ backend-java\
│   └─ target\
│       └─ mars-service-2.0.0.jar  <-- built in step 2.2
├─ frontend\                <-- static React build
├─ config\
│   └─ .env                  <-- leave placeholders; wizard fills this
├─ logs\                    <-- create empty folder if missing
├─ install-service.bat
└─ uninstall-service.bat
```



Do

not

fill in real passwords in `config\.env` on the dev machine. Leave placeholders. The setup wizard on the plant server will write the correct values at runtime.

3

Prepare the Plant Server

Copy files and initialise the PostgreSQL schema

3.1

Copy the application folder to the server

Place the folder at a fixed path with no spaces. Recommended location:

```
C:\MesLineSideBoard\
```

If using a USB drive or network share, copy the entire `Drishti-LineSideBoard` folder to that path.



All paths in `install-service.bat` are relative to the batch file location, so the exact drive letter does not matter — but the path must

not contain spaces

3.2

Initialise the PostgreSQL database schema

Open an **elevated Command Prompt** and run:

```
psql -U lsb_admin -d lsb_oe -f C:\MesLineSideBoard\backend-go\db\schema.sql
```

You will be prompted for the `lsb_admin` password you created in step 1.3. Expected output:

```
CREATE TABLE
CREATE TABLE
CREATE TABLE
```

If you see errors, check that PostgreSQL is running: `sc query postgresql-x64-15`

3.3

Open firewall port 3001 for the dashboard

From an elevated Command Prompt:

```
netsh advfirewall firewall add rule ^
name="LSB Dashboard" ^
dir=in action=allow protocol=TCP localport=3001
```

Shop-floor screens and browsers on the plant network will access the dashboard at `http://<server-ip>:3001`.

4

Install Windows Services

Register LSB-Java and LSB-Go as auto-start Windows services using NSSM

4.1

Confirm NSSM is on the PATH

```
nssm version
```

Expected: *NSSM 2.24 (build ...) 64-bit*. If you get "not recognized", copy `nssm.exe` to `C:\Windows\System32\` and retry.

4.2

Run the installer script

From an **elevated Command Prompt**, navigate to the application folder and run:

```
cd C:\MesLineSideBoard
install-service.bat
```

The script will pause and remind you to fill in `config\.env` — press any key to continue (the wizard will fill it in later). The script installs and starts both services automatically.



The script creates the `logs\` folder automatically if it does not exist. Services are set to **auto-start**

— they will restart after a server reboot.

4.3

Verify both services are running

```
sc query LSB-Java
sc query LSB-Go
```

Both must report:

```
STATE : 4    RUNNING
```

If a service is not running, check the log files:

```
type C:\MesLineSideBoard\logs\lsb-java-error.log
type C:\MesLineSideBoard\logs\lsb-go-error.log
```

4.4

Confirm the health endpoint responds

```
curl http://localhost:3001/health
```

Expected response:

```
{"status":"unconfigured","mode":"unconfigured"}
```

unconfigured is the correct initial state — it means the services are up but no database connections have been made yet. Proceed to Phase 5.



If curl returns *connection refused*, the Go service has not fully started yet. Wait 30 seconds and retry. If it still fails, check `logs\lsb-go-error.log`.

5

Setup Wizard — Step 1: Connect to MARS (MSSQL)

Configure the read-only connection to the MARS SQL Server database

5.1

Open the dashboard in a browser

On the plant server (or any machine on the same network), navigate to:

```
http://<plant-server-ip>:3001
```

Because the application is in *unconfigured* mode, you will be automatically redirected to the Setup Wizard screen.

Enter MARS SQL Server connection details

Fill in the fields on **Setup Step 1 — MARS Database:**

Field	Value to enter
Server / Host	MARS SQL Server hostname or IP address
Port	1433 (default)
Database name	MARS database name (from your DBA)
Username	Read-only SQL login (from your DBA)
Password	SQL login password (from your DBA)



Use only the read-only account.

The LSB application issues only `SELECT` queries against MARS. Never enter `sa` or any account with write permissions.

Click "Test & Save" and verify the result

The wizard will attempt an ODBC connection to the SQL Server. Wait up to 30 seconds for the result.



Success:

Green banner — "MARS connection successful." The wizard advances to Step 2 automatically.



Failure:

Red banner with an error message. Common causes and fixes:

Error message	Likely cause	Fix
[Microsoft][ODBC...] Login failed for user	Wrong username or password	Verify credentials with DBA
[Microsoft][ODBC...] Cannot open server	Wrong hostname / firewall blocking port 1433	Ping the server; check firewall rules with network team
ODBC driver not found	ODBC Driver 18 not installed	Install Microsoft ODBC Driver 18 for SQL Server and restart the LSB-Go service

Setup Wizard — Step 2: Connect to PostgreSQL (OEE Database)

Configure the read-write OEE data store

Enter PostgreSQL connection details

You should now be on **Setup Step 2 — OEE Database**. Fill in:

Field	Value to enter
Host	<code>localhost</code> (PostgreSQL is on the same server)
Port	<code>5432</code> (default)
Database name	<code>lsb_oe</code>
Username	<code>lsb_admin</code>
Password	The password you set in step 1.3

6.2

Click "Test & Save" and verify the result



Success:

Green banner — "PostgreSQL connection successful. Database is ready." The wizard closes and you are redirected to the live dashboard.



Failure — common causes:

Error message	Likely cause	Fix
password authentication failed	Wrong password for <code>lsb_admin</code>	Reset with: <code>ALTER USER lsb_admin WITH PASSWORD 'new-password';</code> in <code>psql</code>
database " <code>lsb_oe</code> " does not exist	Database not created	Repeat step 1.3 to create the database
connection refused	PostgreSQL service not running	<code>net start postgresql-x64-15</code>
relation " <code>oe_entries</code> " does not exist	Schema not initialised	Repeat step 3.2 to run <code>schema.sql</code>

6.3

What the wizard saves automatically

After both steps succeed, the wizard writes all connection details into `config\.env` and transitions the application from *unconfigured* → *production* mode. You do not need to edit any files manually.



`config\.env` now contains your database credentials. Ensure this file has restricted NTFS permissions — only the service account and IT admins should be able to read it.

7.1

Health endpoint — confirm production mode

```
curl http://localhost:3001/health
```

Expected after completing the wizard:

```
{"status":"ok","mode":"production"}
```

7.2

Dashboard verification checklist

Open the dashboard at `http://localhost:3001` and confirm each item:

- ☐ Page title shows "Line Side Board" (not setup wizard)
- ☐ Mode indicator in the footer reads **Production**
- ☐ KPI tiles show real numbers from MARS (not dashes or zeros)
- ☐ Production table populates with station data
- ☐ Quality and schedule widgets load without error
- ☐ OEE entry form can be saved (test with a dummy entry, then delete)
- ☐ Downtime events list loads without error
- ☐ Page is accessible from a different PC on the plant network at `http://<server-ip>:3001`

7.3

Service status — final check

```
sc query LSB-Java
sc query LSB-Go
```

Both must show `STATE : 4 RUNNING`. Record the date and time of successful deployment below for the site log.

SERVICE STATUS

- LSB-Java — RUNNING
- LSB-Go — RUNNING

DATA CONNECTIONS

- MARS MSSQL — connected (read-only)
- PostgreSQL lsb_oeo — connected

7.4

Reboot test (recommended)

Reboot the server and confirm both services start automatically:

```
After reboot:
sc query LSB-Java
sc query LSB-Go
```

Both should be in RUNNING state within 60 seconds of Windows login. If a service does not start, check the error log — the most common cause is a dependency (Java or PostgreSQL) that has not finished starting yet. Increase the `AppRestartDelay` if necessary.

8

Ongoing Operations Reference

Day-to-day commands for IT support staff

8.1

Service management commands

Task	Command (elevated prompt)
Check service status	<code>sc query LSB-Java && sc query LSB-Go</code>
Start both services	<code>net start LSB-Java && net start LSB-Go</code>
Stop both services	<code>net stop LSB-Go && net stop LSB-Java</code>
Restart Go service only	<code>net stop LSB-Go && net start LSB-Go</code>
Remove services completely	<code>uninstall-service.bat</code>

8.2

Log file locations

Log file	Contents
<code>logs\lsb-go.log</code>	Go service — normal operation, request log, DB connections
<code>logs\lsb-go-error.log</code>	Go service — startup errors, panics, ODBC failures
<code>logs\lsb-java.log</code>	Java service — Spring Boot startup, MARS query results
<code>logs\lsb-java-error.log</code>	Java service — SQL exceptions, connection failures

Logs are plain text — open with Notepad, or tail in PowerShell:

```
Get-Content C:\MesLineSideBoard\logs\lsb-go.log -Wait -Tail 50
```

8.3

Updating the application (new version)

When a new build is available from the dev team:

```

1. Stop services
net stop LSB-Go
net stop LSB-Java

2. Replace binaries (do NOT overwrite config\.env)
copy /Y new-build\backend-go\lsb-api.exe C:\MesLineSideBoard\backend-go\
copy /Y new-build\backend-java\target\mars-service-2.0.0.jar ^
  C:\MesLineSideBoard\backend-java\target\

3. Restart in dependency order
net start LSB-Java
Wait 15 seconds for Java to fully initialize
net start LSB-Go

```

⚠ Never overwrite `config\.env` during an update. It contains the production database credentials for this plant. Overwriting it returns the application to *unconfigured* mode.

8.4

Quick troubleshooting decision tree

Symptom	First check	If that's fine, check
Dashboard not loading	<code>sc query LSB-Go</code> — is it running?	Firewall — is port 3001 open?
KPIs show dashes / no data	<code>lsb-java-error.log</code> for SQL errors	Ping MARS server; verify credentials in <code>config\.env</code>
OOE entries not saving	<code>lsb-go-error.log</code> for DB errors	<code>sc query postgresql-x64-15</code> — is it running?
Service keeps restarting	Error log for crash reason	Check Java version: <code>java -version</code> must be 17+
Dashboard shows "unconfigured"	<code>config\.env</code> — were credentials accidentally cleared?	Re-run the setup wizard at <code>http://localhost:3001</code>